as observations so that the first stage of two-stage least squares fits the data perfectly, the IV and OLS estimates are identical. More generally, there is a tradeoff between having too many instruments, overfitting at the first stage, and being biased towards OLS, or having too few instruments, and risking dispersion and extreme estimates. Either way, the asymptotic standard errors on which we routinely rely will not properly indicate the degree of bias or the dispersion.

Nelson and Startz (1990a, 1990b) and Maddala and Jeong (1992) have analyzed the case of a univariate regression where the options are OLS or IV estimation with a single instrument. Their results show that the central tendency of the finite-sample distribution of the IV estimator is biased away from the true value and towards the OLS value. Perhaps most seriously, the asymptotic distribution is a very poor approximation to the finite-sample distribution when the instrument is a poor one, in the sense that it is close to orthogonal to the explanatory variable. Additional evidence of poor performance comes from Bound, Jaeger, and Baker (1993), who show that the empirical results in Angrist and Krueger (1991), who used up to 180 instruments with 30,000 observations, can be closely reproduced with randomly generated instruments. Both sets of results show that poor instruments do not necessarily reveal themselves as large standard errors for the IV estimates. Instead it is easy to produce situations in which y is unrelated to x, and where z is a poor instrument for x, but where the IV estimate of the regression of y on x with z as instrument generates a parameter estimate whose "asymptotic t-value" shows an apparently significant effect. As a result, if IV results are to be credible, it is important to establish first that the instruments do indeed have predictive power for the contaminated right-hand-side variables. This means displaying the first-stage regressions—a practice that is far from routine—or at the least examining and presenting evidence on the explanatory power of the instruments. (Note that when calculating two-stage least squares, the exogenous x variables are also included on the right-hand-side with the instruments, and that it is the predictive power of the latter that must be established, for example, by using an F-test for those variables rather than the R^2 for the regression as a whole.)

In recent work, Staiger and Stock (1993) have proposed a new asymptotic theory for IV when the instruments are only weakly correlated with the regressors, and have produced evidence that their asymptotics provides a good approximation to the finite-sample distribution of IV estimators, even in difficult cases such as those examined by Nelson and Startz. These results may provide a better basis for IV inference in future work.

2.7 Using a time series of cross sections

Although long-running panels are rare in both developed and developing countries, independent cross-sectional household surveys are frequently conducted on a regular basis, sometimes annually, and sometimes less frequently. In Chapter 1, I have already referred to and illustrated from the Surveys of Personal Income Distribution in Taiwan (China), which have been running annually since 1976, and I shall use these data further in this section. Although such surveys select

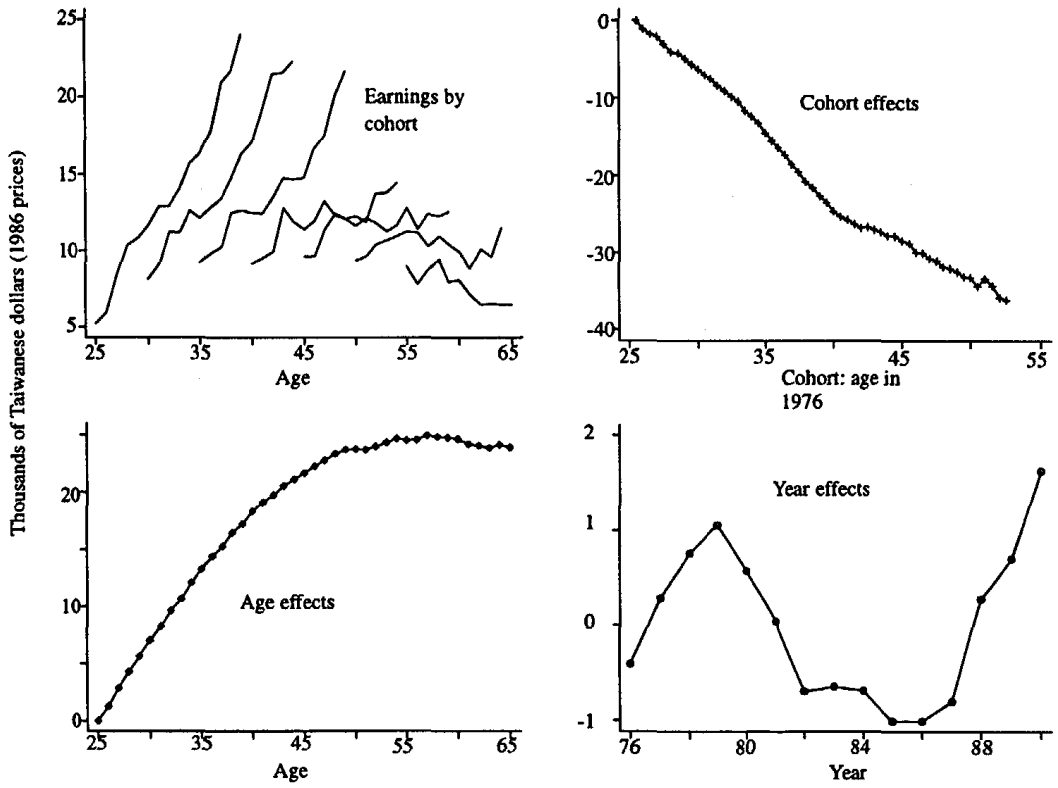
different households in each survey, so that there is no possibility of following *individuals* over time, it is still possible to follow *groups* of people from one survey to another. Obvious examples are the group of the whole population, where we use the surveys to track aggregate data over time, or regional, sectoral, or occupational groups, where we might track the differing fortunes over time of farmers versus government servants, or where we might ask whether poverty is diminishing more rapidly in one region than in another.

Perhaps somewhat less obvious is the use of survey data to follow *cohorts* of individuals over time, where cohorts are defined by date of birth. Provided the population is not much affected by immigration and emigration, and provided the cohort is not so old that its members are dying in significant numbers, we can use successive surveys to follow each cohort over time by looking at the members of the cohort who are randomly selected into each survey. For example, we can look at the average consumption of 30-year-olds in the 1976 survey, of 31-year-olds in the 1977 survey, and so on. These averages, because they relate to the same group of people, have many of the properties of panel data. Cohorts are frequently interesting in their own right, and questions about the gainers and losers from economic development are often conveniently addressed by following such groups over time. Because there are many cohorts alive at one time, cohort data are more diverse and richer than are aggregate data, but their semiaggregated structure provides a link between the microeconomic household-level data and the macroeconomic data from national accounts. The most important measures of living standards, income and consumption, have strong life-cycle age-related components, but the profiles themselves will move upward over time with economic growth as each generation becomes better-off than its predecessors. Tracking different cohorts through successive surveys allows us to disentangle the generational from life-cycle components in income and consumption profiles.

Cohort data: an example

The left-hand top panel of Figure 2.5 shows the averages of real earnings for various cohorts in Taiwan (China) observed from 1976 through to 1990. The data were constructed according to the principles outlined above. For example, for the cohort born in 1941, who were 35 years old in 1976, I used the 1976 survey to calculate the average earnings of all those aged 35, and the result is plotted as the first point in the third line from the left in the figure. The average earnings of 36-year-olds in the 1977 survey is calculated and forms the second point on the same segment. The rest of the line comes from the other surveys, tracking the cohort born in 1941 through the 15 surveys until they are last observed at age 49 in 1990. Table 2.2 shows that there were 699 members of the cohort in the 1976 survey, 624 in the 1977 survey, 879 in the 1978 survey (in which the sample size was increased), and so on until 691 in 1990. The figure illustrates the same process for seven cohorts, born in 1951, 1946, and so on backward at five-year intervals until the oldest, which was born in 1921, and the members of which were 69 years old when last seen in 1990. Although it is possible to make graphs for all

Figure 2.5. Earnings by cohort and their decomposition, Taiwan (China), 1976–90



Note: Author's calculations based on Surveys of Personal Income Distribution.

Table 2.2. Number of persons in selected cohorts by survey year, Taiwan (China), 1976–90

	Cohort: age in 1976						
Year	*25*	*30*	*35*	*40*	*45*	*50*	*55*
1976	863	521	699	609	552	461	333
1977	902	604	624	535	585	427	308
1978	1,389	854	879	738	714	629	477
1979	1,351	796	846	708	714	574	462
1980	1,402	834	845	723	746	625	460
1981	1,460	794	807	720	750	624	426
1982	1,461	771	838	695	689	655	496
1983	1,426	737	846	718	702	597	463
1984	1,477	825	820	711	695	541	454
1985	1,396	766	775	651	617	596	442
1986	1,381	725	713	659	664	549	428
1987	1,309	634	775	632	675	513	0
1988	1,275	674	700	617	595	548	0
1989	1,225	672	652	600	609	519	0
1990	1,121	601	691	575	564	508	0

Note: The year is the year of the survey, and the numbers are the numbers of individuals in each cohort sampled in each survey year. 65 is used as an age cutoff, so the oldest cohort is not observed after 1986.
Source: Author's calculations from the Surveys of Personal Income Distribution.

birth years, I have shown only every fifth cohort so as to keep the diagram clear. Note that only members of the same cohort are joined up by connecting lines, and this construction makes it clear when we are following different groups of people or jumping from one cohort to another. (See also Figures 6.3 and 6.4 below for the corresponding graphs for consumption and for a comparison of cross-sectional and cohort plots.)

The top left-hand panel of the figure shows clear age and cohort effects in earnings; it is also possible to detect common macroeconomic patterns for all cohorts. With a very few exceptions at older ages, the lines for the younger cohorts are always above the lines for the older cohorts, even when they are observed at the same age. This is because rapid economic growth in Taiwan (China) is making younger generations better-off, so that, for example, those born in 1951—the youngest, left-most cohort in the figure—have average earnings at age 38 that are approximately twice as much as the earnings at age 38 of the cohort born 10 years earlier—the third cohort in the figure. There is also a pronounced life-cycle profile to earnings, and although the age profile is "broken up" by the cohort effects, it is clear that earnings tend to grow much more rapidly in the early years of the working life than they do after age 50. As a result, not only are the younger cohorts of workers in Taiwan (China) better-off than their predecessors, but they have also experienced much more rapid growth in earnings. The macroeconomic effects in the first panel of Figure 2.5 are perhaps the hardest to see, but note that each connected line segment corresponds to the same contemporaneous span of 15 years in "real" time, 1976–90. Each segment shows the impact

of the slowdown in Taiwanese economic growth after the 1979 oil shock. Each cohort has very rapid growth from the second to third year observed, which is 1977–78, somewhat slower growth for the next two years, 1978–80, and then two years of slow or negative growth after the shock. This decomposition into cohort, age, and year effects can be formalized in a way that will work even when the data are not annual and not necessarily evenly spaced, a topic to which I return in the final subsection below. Before that, however, it is useful to use this example to highlight the advantages and disadvantages of cohort data more generally.

Cohort data versus panel data

A useful comparison is between the semiaggregated cohort data and genuine panel data in which individual households are tracked over time. In both cases, we have a time series of observations on a number of units, with units defined as either cohorts or individuals. The cohort data cannot tell us anything about dynamics within the cohorts; each survey tells us about the distribution of the characteristic in the cohort in each period, but two adjacent surveys tell us nothing about the joint distribution of the characteristic in the two periods. In the earnings example, the time series of cross sections can tell us about average earnings for the cohort over time, and it can tell us about inequality of earnings within the cohort and how it is changing over time, but it cannot tell us how long individuals are poor, or whether the people who are rich now were rich or poor at some earlier date. But apart from dynamics, the cohort data can do most of what would be expected of panel data. In particular, and as we shall see in the next subsection, the cohort data can be used to control for unobservable fixed effects just as with panel data, a feature that is often thought to be the main econometric attraction of the latter.

Cohort data also have a number of advantages over most panels. As we have seen in Chapter 1, many panels suffer from attrition, especially in the early years, and so run the risk of becoming increasingly unrepresentative over time. Because the cohort data are constructed from fresh samples every year, there is no attrition. There will be (related) problems with the cohort data if the sampling design changes over time, or if the probabilities of selection into the sample depend on age as, for example, for young men undergoing military training. The way in which the cohort data are used will often be less susceptible to measurement error than is the case with panels. The quantity that is being tracked over time is typically an average (or some other statistic such as the median or other percentile) and the averaging will nearly always reduce the effects of measurement error and enhance the signal-to-noise ratio. In this sense, cohort methods can be regarded as IV methods, where the instruments are grouping variables, whose application averages away the measurement error. Working with aggregated data at a level that is intermediate between micro and macro also brings out the relationship between household behavior and the national aggregates and helps bridge the gap between them; in Figure 2.5, for example, the behavior of the aggregate economy is clearly apparent in the averages of the household data.

It should be emphasized that cohort data can be constructed for any characteristic of the distribution of interest; we are not confined to means. As we shall see in Chapter 6, it can be interesting and useful to study how inequality changes within cohorts over time, and since we have the micro data for each cohort in each year, it is as straightforward to work with measures of dispersion as it is to work with measures of central tendency. Medians can be used instead of means—a technique that is often useful in the presence of outliers—and if the theory suggests working with some transform of the data, the transform can be made prior to averaging. When working with aggregate data, theoretical considerations often suggest working with the mean of a logarithm, for example, rather than with the logarithm of the mean. The former is not available from aggregate data, but can be routinely computed from the micro data when calculating the semiaggregated cohort averages.

A final advantage of cohort methods is that they allow the combination of data from different surveys on different households. The means of cohort consumption from an expenditure survey can be combined with the means of cohort income from a labor force survey, and the hybrid data set used to study saving. It is not necessary that all variables are collected from the same households in one survey.

Against the use of cohort data, it should be noted that there are sometimes problems with the assumption that the cohort population is constant, an assumption that is needed if the successive surveys are to generate random samples from the same underlying population. I have already noted potential problems with military service, migration, aging, and death. But the more serious difficulties come when we are forced to work, not with individuals, but with households, and to define cohorts of households by the age of the head. If households once formed are indissoluble, there would be no difficulty, but divorce and remarriage reorganize households, as does the process whereby older people go to live with their children, so that previously "old" households become "young" households in subsequent years. It is usually clear when these problems are serious, and they affect some segments of the population more than others, so that we know which data to trust and which to suspect.

Panel data from successive cross sections

It is useful to consider briefly the issues that arise when using cohort data as if they were repeated observations on individual units. I show first how fixed effects at the individual level carry through to the cohort data, and what steps have to be taken if they are to be eliminated. Consider the simplest univariate model with fixed effects, so that at the level of the individual household, we have (2.76) with a single variable

$$y_{it} = \alpha + \beta x_{it} + \mu_t + \theta_i + u_{it} \tag{2.84}$$

where the μ_t are year dummies and θ_i is an individual-specific fixed effect. If there were no fixed effects, it would be possible to average (2.84) over all the

households in each cohort in each year to give a corresponding equation for the cohort averages. When there are fixed effects, (2.84) still holds for the cohort population means, with cohort fixed effects replacing the individual fixed effects. However, if we average (2.84) over the members of the cohorts who appear in the survey, and who will be different from year to year, the "fixed effect" will not be fixed, because it is the average of the fixed effects of different households in each year. Because of this sampling effect, we cannot remove the cohort fixed effects by differencing or using within-estimators.

Consider an alternative approach based on the unobservable population means for each cohort. Start from the cohort version of (2.84), and denote population means in cohorts by the subscripts c, so that, simply changing the subscript i to c, we have

$$y_{ct} = \alpha + \beta x_{ct} + \mu_t + \theta_c + u_{ct} \tag{2.85}$$

and take first differences—the comparable analysis for the within-estimator is left as an exercise—to eliminate the fixed effects so that

$$\Delta y_{ct} = \Delta \mu_t + \beta \Delta x_{ct} + \Delta u_{ct} \tag{2.86}$$

where the first term is a constant in any given year. This procedure has eliminated the fixed effects, but we are left with the unobservable changes in the population cohort means in place of the sample cohort means, which is what we observe. If we replace Δy and Δx in (2.84) by the observed changes in the sample means, we generate an error-in-variables problem, and the estimates will be attenuated.

There are at least two ways of dealing with this problem. The first is to note that, just as the sample was used to provide an estimate of the cohort mean, it can also be used to provide an estimate of the standard error of the estimate, which in this context is the variance of the measurement error. For the example (2.86), we can use overbars to denote sample means and write

$$\begin{aligned} \Delta \bar{y}_{ct} &= \Delta y_{ct} + \epsilon_{1ct} - \epsilon_{1ct-1} \\ \Delta \bar{x}_{ct} &= \Delta x_{ct} + \epsilon_{2ct} - \epsilon_{2ct-1} \end{aligned} \tag{2.87}$$

where ϵ_{1ct} and ϵ_{2ct} are sampling errors in the cohort means. Because they come from different surveys with independently selected samples, they are independent over time, and their variances and covariance, σ_1^2, σ_2^2, and σ_{12} are calculated in the usual way, from the variances and covariance in the sample divided by the cohort size (with correction for cluster effects as necessary.) From (2.87), we see that the variances and covariances of the sample cohort means are inflated by the variances and covariances of the sampling errors, but that, if these are subtracted out, we can obtain consistent estimates of β in (2.86) from—cf. also (2.62) above,

$$\tilde{\beta} = \frac{\operatorname{cov}(\Delta \bar{x}_{ct} \Delta \bar{y}_{ct}) - \sigma_{12t} - \sigma_{12t-1}}{\operatorname{var}(\Delta \bar{x}_{ct}) - \sigma_{2t}^2 - \sigma_{2t-1}^2} \tag{2.88}$$

and where, for illustrative purposes, I have assumed that there are only two time periods t and $t-1$. The standard error for (2.88) can be calculated using the bootstrap or the delta method—discussed in the next section—which can also take

into account the fact that the variance and covariances of the sampling errors are estimated (see Deaton 1985, who also discusses the multivariate case, and Fuller 1987, who gives a general treatment for a range of similar models).

Another possible estimation strategy is to use IV, with changes from earlier years used as instruments. Since the successive samples are independently drawn, changes in cohort means from $t-2$ to $t-1$ are measured independently of the change from t to $t+1$. In some cases, the cohort samples may be large enough and the means precisely enough estimated so that these corrections are small enough to ignore. In any case, it is a good idea to check the standard errors of the cohort means, to make sure that regression results are not being dominated by sampling effects, and if so, to increase the cohort sizes, for example, by working with five-year age bands instead of single years. In some applications, this might be desirable on other grounds; in some countries, people do not know their dates of birth well enough to be able to report age accurately, and reported ages "heap" at numbers ending in 5 and 0.

Decompositions by age, cohort, and year

A number of the quantities most closely associated with welfare, including family size, earnings, income, and consumption, have distinct and characteristic life-cycle profiles. Wage rates, earnings, and saving usually have hump-shaped age profiles, rising to their maximum in the middle years of life, and declining somewhat thereafter. The natural process of bearing and raising children induces a similar profile in average family size. Moreover, all of these quantities are subject to secular variation; consumption, earnings, and incomes rise over time with economic development, and family size decreases as countries pass through the demographic transition. In consequence, even if the shape of the age profiles remains the same for successive generations, their position will shift from one to the next. The age profile from a single cross section confounds the age profile with the generational or cohort effects. For example, a cross-sectional earnings profile will tend to exaggerate the downturn in earnings at the highest age because, as we look at older and older individuals, we are not just moving along a given age-earnings profile, but we are also moving to ever lower lifetime profiles. The cohort data described in this section allow us to track the same cohort over several years and thus to avoid the difficulty; indeed, the Taiwanese earnings example in Figure 2.5 provides a clear example of the differences between the age profiles of different cohorts. In many cases, diagrams like Figure 2.5 will tell us all that we need to know. However, since each cohort is only observed for a limited period of time, it is useful to have a technique for linking together the age profiles from different cohorts to generate a single complete life-cycle age profile. This is particularly true when there is only a limited number of surveys, and the intervals between them are more than one year. In such cases, diagrams like Figure 2.5 are harder to draw, and a good deal less informative.

In this subsection, I discuss how the cohort data can be decomposed into age effects, cohort effects, and year effects, the first to give the typical age profile, the

second the secular trends that lead to differences in the positions of age profiles for different cohorts, and the third the aggregate effects that synchronously but temporarily move all cohorts off their profiles. These decompositions are based on models and are certainly not free of structural assumptions; they assume away interaction effects between age, cohort, and years, so that, for example, the shape of age profiles is unaffected by changes in their position, and the appropriateness and usefulness of the assumption has to be judged on a case-by-case basis.

To make the analysis concrete, consider the case of the lifetime consumption profile. If the growth in living standards acts so as to move up the consumption-age profiles proportionately, it makes sense to work in logarithms, and to write the logarithm of consumption as

$$\ln c_{ct} = \beta + \alpha_a + \gamma_c + \psi_t + u_{ct} \tag{2.89}$$

where the superscripts c and t (as usual) refer to cohort and time (year), and a refers to age, defined here as the age of cohort c in year t. In this particular case, (2.89) can be given a theoretical interpretation, since according to life-cycle theory under certainty, consumption is the product of lifetime wealth, the cohort aggregate of which is constant over time, and an age effect, which is determined by preferences (see Section 6.1 below). In other contexts where there is no such theory, the decomposition is often a useful descriptive device, as for earnings in Taiwan (China), where it is hard to look at the top left-hand panel of Figure 2.5 without thinking about an age and cohort decomposition.

In order to implement a model like (2.89), we need to decide how to label cohorts. A convenient way to do so is to choose c as the age in year $t=0$. By this, c is just a number like a and t. We can then choose to restrict the age, cohort, and year effects in (2.89) in various different ways. In particular, we can choose polynomials or dummies. For the year effects, where there is no obvious pattern *a priori*, dummy variables would seem to be necessary, but age effects could reasonably be modeled as a cubic, quartic, or quintic polynomial in age, and cohort effects, which are likely to be trend-like, might even be adequately handled as linear in c. Given the way in which we have defined cohorts, with bigger values of c corresponding to older cohorts, we would expect γ_c to be declining with c. When data are plentiful, as in the Taiwanese case, there is no reason not to use dummy variables for all three sets of effects, and thus to allow the data to choose any pattern.

Suppose that A is a matrix of age dummies, C a matrix of cohort dummies, and Y a matrix of year dummies. The cohort data are arranged as cohort-year pairs, with each "observation" corresponding to a single cohort in a specific year. If there are m such cohort-year pairs, the three matrices will each have m rows; the number of columns will be the number of ages (or age groups), the number of cohorts, and the number of years, respectively. The model (2.89) can then be written in the form

$$y = \beta + A\alpha + C\gamma + Y\psi + u \tag{2.90}$$

where y is the stacked vector of cohort-year observations—each row corresponds to a single observation on a cohort—on the cohort means of the logarithm of consumption. As usual, we must drop one column from each of the three matrices, since for the full matrices, the sum of the columns is a column of ones, which is already included as the constant term.

However, even having dropped these columns, it is still impossible to estimate (2.90) because there is an additional linear relationship across the three matrices. The problem lies in the fact that if we know the date, and we know when a cohort was born, then we can infer the cohort's age. Indeed, since c is the age of the cohort in year 0, we have

$$a_{ct} = c + t \tag{2.91}$$

which implies that the matrices of dummies satisfy

$$As_a = Cs_c + Ys_y \tag{2.92}$$

where the s vectors are arithmetic sequences $\{0,1,2,3, \ldots, \}$ of the length given by the number of columns of the matrix that premultiplies them. Equation (2.92) is a single identity, so that to estimate the model it is necessary to drop one more column from any one of the three matrices.

The normalization of age, cohort, and year effects has been discussed in different contexts by a number of authors, particularly Hall (1971), who provides an admirably clear account in the context of embodied and disembodied technical progress for different vintages of pickup trucks, and by Weiss and Lillard (1978), who are concerned with age, vintage, and time effects in the earnings of scientists. The treatment here is similar to Hall's, but is based on that given in Deaton and Paxson (1994a). Note first that in (2.90), we can replace the parameter vectors α, γ, and ψ by

$$\tilde{\alpha} = \alpha + \kappa s_a, \quad \tilde{\gamma} = \gamma - \kappa s_c, \quad \tilde{\psi} = \psi - \kappa s_y \tag{2.93}$$

for any scalar constant κ, and by (2.92) there will be no change in the predicted value of y in (2.87). According to (2.90), a time-trend can be added to the age dummies, and the effects offset by subtracting time-trends from the cohort dummies and the year dummies.

Since these transformations are a little hard to visualize, and a good deal more complicated than more familiar dummy-variable normalizations, it is worth considering examples. Suppose first that consumption is constant over cohorts, ages, and years, so that the curves in Figure 2.5 degenerate to a single straight line with slope 0. Then we could "decompose" this into a positive age effect, with consumption growing at (say) five percent for each year of age, and offset this by a negative year effect of five percent a year. According to this, each cohort would get a five percent age bonus each year, but would lose it to a macroeconomic effect whereby everyone gets five percent less than in the previous year. If this

were all, younger cohorts would get less than older cohorts at the same age, because they come along later in time. To offset this, we need to give each cohort five percent more than the cohort born the year previously which, since the older cohorts have higher cohort numbers, means a negative trend in the cohort effects. More realistically, suppose that when we draw Figure 2.5, we find that the consumption of each cohort is growing at three percent a year, and that each successive cohort's profile is three percent higher than that of its predecessor. Everyone gets three percent more a year as they age, and starting consumption rises by three percent a year. This situation can be represented (exactly) by age effects that rise linearly with age added to cohort effects that fall linearly with age by the same amount each year; note that cohorts are labeled by age at a fixed date, so that older cohorts (larger c) are poorer, not richer. But the same data can be represented by a time-trend of three percent a year in the age effects, without either cohort or year effects.

In practice, we choose a normalization that is most suitable for the problem at hand, attributing time-trends to year effects, or to matching age and cohort effects. In the example here, where consumption or earnings is the variable to be decomposed, a simple method of presentation is to attribute growth to age and cohort effects, and to use the year effects to capture cyclical fluctuations or business-cycle effects that average to zero over the long run. A normalization that accomplishes this makes the year effects orthogonal to a time-trend, so that, using the same notation as above,

$$s_y'\psi = 0. \tag{2.94}$$

The simplest way to estimate (2.90) subject to the normalization (2.94) is to regress y on (a) dummies for each cohort excluding (say) the first, (b) dummies for each age excluding the first, and (c) a set of $T-2$ year dummies defined as follows, from $t = 3,..,T$

$$d_t^* = d_t - [(t-1)d_2 - (t-2)d_1] \tag{2.95}$$

where d_t is the usual year dummy, equal to 1 if the year is t and 0 otherwise. This procedure enforces the restriction (2.94) as well as the restriction that the year dummies add to zero. The coefficients of the d_i^* give the third through final year coefficients; the first and second can be recovered from the fact that all year effects add to zero and satisfy (2.94).

This procedure is dangerous when there are few surveys, where it is difficult to separate trends from transitory shocks. In the extreme case where there are only two years, the method would attribute any increase in consumption between the first and second years to an increasing age profile combined with growth from older to younger cohorts. Only when there are sufficient years for trend and cycle to be separated can we make the decomposition with any confidence.

The three remaining panels of Figure 2.5 show the decomposition of the earnings averages into age, cohort, and year dummies. The cohort effects in the top

right-hand panel are declining with age; the earlier you are born, the older you are in 1976, and age in 1976 is the cohort measure. Although the picture is one that is close to steady growth from cohort to cohort, there has been a perceptible acceleration in the rate of growth for the younger cohorts. The bottom left-hand panel shows the estimated age effects; according to this, wages are a concave function of age, and although there is little wage increase after age 50, there is no clear turning down of the profile. Although the top left panel creates an impression of a hump-shaped age profile of earnings, much of the impression comes from the cohort effects, not the age effects, and although the oldest cohort shown has declining wages from ages 58 through 65, other cohorts observed at the same ages do not display the same pattern. (Note that only every fifth cohort is included in the top left panel, but all cohorts are included in the regressions, subject only to age lying being between 25 and 65 inclusive.) The final panel shows the year effects, which are estimated to be much smaller in magnitude than either the cohort or age effects; nevertheless they show a distinctive pattern, with the economy growing much faster than trend at the beginning and end of the period, and much more slowly in the middle after the 1979 oil shock.

Age and cohort profiles such as those in Figure 2.5 provide the material for examining the structural consequences of changes in the rates of growth of population and real income. For example, if the age profiles of consumption and income are determined by tastes and technology, and are invariant to changes in the rate of economic growth, we can change the cohort effects holding the age effects constant and thus derive the effects of growth on aggregates of consumption, saving, and income. Changes in population growth rates redistribute the population over the various ages, so that, once again, we can use the age profiles as the basis for aggregating over different age distributions of the population. Much previous work has been forced to rely on single cross sections to estimate age profiles, and while this is sometimes the best that can be done, cross-sectional age profiles confuse the cohort and age effects, and will typically give much less reliable estimates than the methods discussed in this section. I return to these techniques in the final chapter when I come to examine household saving behavior.

2.8 Two issues in statistical inference

This final section deals briefly with two topics that will be required at various points in the rest of the book, but which do not fit easily into the rest of this chapter. The first deals with a situation that often arises in practice, when the parameters of interest are not the parameters that are estimated, but functions of them. I briefly explain the "delta" method which allows us to transform the variance-covariance matrix of the estimated parameters into the variance-covariance matrix of the parameters of interest, so that we can construct hypothesis tests for the latter. Even when we want to use the bootstrap to generate confidence intervals, asymptotic approximation to variances are useful starting points that can be improved using the bootstrap (see Section 1.4). The second topic is concerned with sample size, and its effects on statistical inference. Applied econometricians often